

# A Comparative Review of Deep Learning Models for Deepfake Detection

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**Abstract**—The development of generative AI has advanced significantly over the past few decades, enabling the creation of deepfake images that are increasingly difficult to distinguish from genuine photographs. The widespread availability of these models, which can be easily used, poses a substantial risk of spreading disinformation. Consequently, there is a pressing need for robust and reliable methods to identify images that have been created or altered using generative AI models. To address this need, a diverse range of methods and models have been developed. However, these approaches are often not exhaustively compared to one another, nor are they evaluated using a common reference dataset. Moreover, the majority of existing deepfake detection models rely on deep learning techniques, with numerous models available for feature extraction and detection, ranging from simple Convolutional Neural Networks (CNNs) to more advanced Vision Transformers (ViTs). To ensure the comparability and reproducibility of deepfake detection models, a standardized benchmark is urgently required to evaluate their performance across a large-scale, common dataset. Such a reference benchmark would facilitate the development of more effective and robust detection methods by providing insight into the strengths and weaknesses of existing AI-based approaches. In addition to establishing this benchmark, this paper also explores the challenges of combining different AI-based deepfake detection models and investigates various aggregation methods to further improve overall detection performance. A large-scale experiment involving almost 50 generative AI methods and over 40 deep learning-based feature extraction and detection models demonstrates the relevance of this study.

**Index Terms**—Deepfake detection, Generative AI, Survey, Review, Benchmarking, Empirical evaluation, Artificial Intelligence, Image Forensics, Media Security

## I. INTRODUCTION

Thanks to advances in artificial intelligence, AI-based image generation tools, including Deep Fakes, are now widely available to the public. These AI-generated images are so sophisticated that they can no longer be distinguished from natural photographs with the naked eye. Advanced statistical methods, particularly those based on Deep Learning, are required to detect them, but even these methods are not always accurate.

We've all seen fake images used to manipulate news on social media, and sometimes even on reputable news websites. This raises significant concerns about the credibility

of digital photographs, which are often used as evidence to support factual claims. This issue is now widely recognized by policymakers, as highlighted in a report by the National Science and Technology Council [1]. As noted in a recent study [2], the use of AI-generated images poses a particular threat in the context of foreign influence, as it enables the creation of "seemingly-authentic and tailored messaging."

To address these risks, researchers have developed various methods to detect images generated by AI [3]–[5]. However, most of these methods rely on Deep Learning, which, while effective, presents a challenge from a forensic perspective. This is because "black box methods" like Deep Learning and AI can lead to models that are difficult to explain and interpret [6]. To improve media forensic investigations, it's essential to enhance the traceability of AI decisions and develop more interpretable models."

## A. Position and Contribution of Present Paper

Recent deepfake detection methods have predominantly employed deep learning techniques, with a vast majority of approaches relying on these methods to identify and mitigate the spread of manipulated media. Even methods that focus on exploiting specific features, such as the upsampling operation of images generated by generative AI, often utilize a deep learning model as a detector. For instance, a custom CNN model was used in the method proposed by Nataraj et al. [7] to detect the specific upsampling operation, while Barni et al. [8] employed an ad hoc CNN architecture to exploit the co-occurrence in spectral subband.

Reference and well-know deep learning models have been extensively used to address the problem of deepfake detection. For instance, the VGG network was used by Chang et al. [9] for detection of face images generated using GAN-based generative AI. Similarly, Inception-ResNet-v2 was employed by Rajalaxmi et al. [10] for similar purposes of deepfake face detection. Other deep learning architectures, such as ResNet50 and XceptionNet, have also been used for deepfake detection, as, for example, in the work of Coccomini et al. [11] and Wang et al. [12], [13]. EfficientNetB4, one of the variants of the EfficientNet architecture, has been used for fake video detection, image detection generated by GANs, and detection of images generated by diffusion models, as demonstrated in

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the work of Bonettini et al. [14], Mandelli et al. [15], and Mandelli et al. [16], respectively.

The recent introduction of Vision Transformers has revolutionized the field of deep learning-based detection and feature extraction, and has been quickly adopted for deepfake detection, as, for instance, in the work of Wodajo et al. [17] which proposed a custom model. Additionally, an approach combining EfficientNet and transformers with an attention-based mechanism was leveraged by Coccomini et al. [18] for the detection of video deepfakes. Other transformer-based deep learning architectures, such as MobileNetv2 block (MNV2) [19] for the detection of deepfake images generated by diffusion-based models. Similarly, RegNet was used in [20] for Deepfake detection with applications on face manipulation and maxViT was used in conjunction with ResNet in [21] for exposing deepfakes in general. More recently, a Challenge on DeepFake Analysis and Detection (DFAD 2023) has been organized during the International Conference on Computer Vision (ICCV 2023) in Paris and the winner of this challenge used the well-celebrated swin transformer, as originally proposed in [22], [23].

From the previous quick review of the state-of-the-art, one can conclude that that deep learning-based method has been extensively used for all possible applications of deepfake detection. This trend raises fundamental questions, for instance, about the generalizability and robustness of these approaches and highlights the fundamental need for a more comprehensive comparison of these models in the context of deepfake detection. Indeed, a thorough analysis of the state-of-the-art and detailed comparison of these models, including their strengths, weaknesses, and limitations. Moreover, the lack of a general comparison of deep learning for deepfake detection in general is striking. Most of the works we cited above are based on their own dataset of a very limited number of generative AI models. A more comprehensive evaluation of deep learning models against different deepfake generation models would enable researchers to identify the most effective models and would possibly provide valuable insights into the most effective approaches for deepfake detection. Ultimately, the lack of a reference benchmark and comparison framework prevents the development of more effective and robust deepfake detection systems, and highlights the need for a more systematic and comprehensive evaluation of deep learning models in this context.

## II. PRACTICAL METHODOLOGY AND EXPERIMENTAL FRAMEWORK

In our experimentation we generated 12,000 images from a wide range of generative AI models. For a numerical comparison of the performance of a deep learning-based method for deepfake detection, our goal was to create a dataset of deepfake images as large as possible. In total we ended up with a total of 48 classes of image based on 47 generative AI models plus one real photograph. For the real photograph, we used a mix of ALASKA image datasets [24], [25] and images collected from the FlickrR photo-sharing platform and from

the pexels free stock images website. The total number of real photographs was 200,000 in the training set and 20,000 in the testing set. The list of text-to-image generative models we used is detailed in the Table I. We have tried to be as exhaustive as possible, including the first GAN-based image generator, such as Glide or Lafite, and the latest diffusion-transformer based image generator, such as stable diffusion 3.5 and FLUX.1. We used the generative AI models as proposed on the Huggingface website (see Table I for the precise reference for all generators). The images were generated using a random choice from the One Million Random Midjourney Prompts which were collected from the Discord channel interface for Midjourney. The guidance scales and number of diffusion steps were also randomized and all the other settings of the generator were left to default. For each text-to-image generator, we used 10,000 images for training and 2,000 images for testing.

Regarding the classification, all the models we used were obtained from the library timm [26] from which we loaded models pretrained over the ImageNet dataset because this generally greatly improves training convergence. All the models were trained using a decreasing learning rate following stochastic gradient descent with warm restarts (SGDR) [27]–[29]. Given the size of the training dataset with 670,000 images, we used “only” 35 epochs. The initial value for the learning rate was obtained using from the method initially proposed in [30]. In brief, it essentially consists in a 1-cycle training of the deep-learning method over so-called mini-batches : the learning rate is gradually increased, at each mini-batch, from a very low initial value to a final high value. Throughout this process, the loss is measured at each iteration in order to find the largest value, with a margin value before the loss begins to diverge.

The learning rate scheduler is thus based on this initial guess and then slowly decreased over one cycle. The initial cycle length is set to 5, it is doubled for every cycle and the initial learning rate is divided by two after each cycle. We used three cycles for a total of 35 epochs, which is far enough for convergence, as shown in the next Section III and especially in the Figure 1.

Another important factor that we noticed in the importance of data augmentation to prevent overfitting of the model and ensure a better generalization even though testing and training sets were generated in the very same manner ; they are, in fact, a random split from the same dataset. This fact has also been reported in [18], [31]. In our case we carried out data augmentation by adding the following operation : mirroring, flipping along x-axis and y-axis, Gaussian i.i.d. noise addition (with standard deviation between 0.05 and 0.15), multiplication noise addition (with a factor between 0.975 and 1.025) resizing (with a rescaling factor between 0.95 and 1.05) and rotation (with an angle between -5 and 5 degrees). Each operation was applied independently, using the Albumentations library, and with a probability  $p = 0.25$  for each. While each operation within the overall data augmentation process does not modify the image significantly, and, from a quick search, we did not find interest in applying more important operations, we have

| Type                    | Name                                  | URL / Source |
|-------------------------|---------------------------------------|--------------|
| Transformers            | FLUX.1-dev                            | HuggingFace  |
|                         | FLUX.1-schnell                        | HuggingFace  |
|                         | Lumina-T2I                            | HuggingFace  |
|                         | Unidiffuser-v1                        | HuggingFace  |
|                         | Stable Diffusion 3                    | HuggingFace  |
|                         | Stable Diffusion 3.5                  | HuggingFace  |
| Diffusion Models        | Kandinsky v2.1                        | HuggingFace  |
|                         | Kandinsky v2.2                        | HuggingFace  |
|                         | Kandinsky v3                          | HuggingFace  |
|                         | Pixart- $\alpha$                      | HuggingFace  |
|                         | Pixart- $\Sigma$                      | HuggingFace  |
|                         | DreamLike 2.0                         | HuggingFace  |
|                         | Playground 2.0                        | HuggingFace  |
|                         | Playground 2.5                        | HuggingFace  |
|                         | Stable Diffusion 1.5                  | HuggingFace  |
|                         | Stable Diffusion 2.1                  | HuggingFace  |
|                         | Stable Diffusion XL                   | HuggingFace  |
|                         | Stable Diffusion XL-Turbo             | HuggingFace  |
|                         | Stable Diffusion Refiner              | HuggingFace  |
|                         | Stable Cascade                        | HuggingFace  |
|                         | CogView 3                             | HuggingFace  |
|                         | ConrolNetXS                           | HuggingFace  |
|                         | Kolors                                | HuggingFace  |
|                         | Wuerstchen v2                         | HuggingFace  |
|                         | DeepFloyd-IF                          | HuggingFace  |
|                         | Latent-Diff                           | GitHub       |
|                         | ShiftedDiff                           | GitHub       |
| GANs                    | StyleGAN3                             | GitHub       |
|                         | StyleGAN2                             | GitHub       |
|                         | GigaGAN                               | GitHub       |
|                         | Glide                                 | GitHub       |
|                         | Lafite                                | GitHub       |
|                         | Dall-E Mini                           | HuggingFace  |
|                         | Dall-E Mega                           | HuggingFace  |
| Finetuning / Variations | Stable-Diff Turbo (from SD 2.1)       | HuggingFace  |
|                         | animagine XL 3.0 (from SDXL)          | HuggingFace  |
|                         | lcm-LoRa SDXL Turbo                   | HuggingFace  |
|                         | Fire Generation (from SDXL-Turbo)     | HuggingFace  |
|                         | DreamBooth (from SDXL-Turbo)          | HuggingFace  |
|                         | SDXL Turbo DPO LoRA (from SDXL-Turbo) | HuggingFace  |
|                         | Pixart- $\alpha$ -LCM                 | HuggingFace  |
|                         | Pixart- $\alpha$ -ControlNet          | HuggingFace  |
|                         | Pixart- $\alpha$ -DMD                 | HuggingFace  |
|                         | Kolors-PAG                            | HuggingFace  |
|                         | Jovie-Midjourney (from FLUX)          | HuggingFace  |
|                         | flux-schnell-realism (from FLUX)      | HuggingFace  |
|                         | Jovie-Retro (from FLUX)               | HuggingFace  |

TABLE I: List of generators and their accessibility.

found that these processing were enough to greatly improve the testing accuracy.

#### A. List of Deep Learning Models for Deepfake Detection

The list of all 41 models we used for deepfake detection is given in the table III along with the year it was released, the link to the arxiv publication and the number of trainable parameters, which give a rough idea of the models complexity. Without describing in detail the architecture of all these models, we would like to briefly justify the choice behind this selection. The family of Efficient Net models was included because it is widely recognized as a reference architecture

and because it has often been used in the field of deepfake detection [14]–[16]. The idea behind efficient net is to find a tradeoff between increases in the depth of the network, its resolution and the width of the architecture and to jointly optimize those parameters.

We also included NFNNet and Mix-Net because these two are somewhat derivations from the EfficientNet approach and because, given the popularity of these models, it is unsurprising that they have been used for deepfake detection [32].

Similarly, we also used several reference and well-known CNN models, such as VGG, DenseNet, ResNet and Incep-

| Name                | year | URL (arXiv) | # Parameters |
|---------------------|------|-------------|--------------|
| EfficientNet2 m     | 2021 | arXiv Link  | ~ 20.2 M     |
| EfficientNet2 s     | 2021 | arXiv Link  | ~ 52.9 M     |
| MixNet              | 2019 | arXiv Link  | ~ 5.8 M      |
| NFNet               | 2021 | arXiv Link  | ~ 32.8 M     |
| TinyNet             | 2020 | arXiv Link  | ~ 4.9 M      |
| ConvNeXt-V2         | 2023 | arXiv Link  | ~ 87.7 M     |
| ResNeXt             | 2016 | arXiv Link  | ~ 192 M      |
| ResNetClip 101      | 2021 | arXiv Link  | ~ 42.6 M     |
| Res2NeXt 50         | 2019 | arXiv Link  | ~ 22.7 M     |
| ResNet 101          | 2015 | arXiv Link  | ~ 42.6 M     |
| SEResNext           | 2021 | arXiv Link  | ~ 91.6 M     |
| VGG19               | 2014 | arXiv Link  | ~ 139 M      |
| DenseNet            | 2016 | arXiv Link  | ~ 18.1 M     |
| Xception            | 2016 | arXiv Link  | ~ 40.4 M     |
| Inception-V3        | 2015 | arXiv Link  | ~ 21.9 M     |
| Inception-ResNet-V2 | 2016 | arXiv Link  | ~ 54.4 M     |
| Lambda ResNet50     | 2021 | arXiv Link  | ~ 19.6 M     |
| MobileNet-V2        | 2018 | arXiv Link  | ~ 4.4 M      |
| MobileNet-V3        | 2019 | arXiv Link  | ~ 4.2 M      |
| MobileNet-V4        | 2024 | arXiv Link  | ~ 3.6 M      |
| MobileViT-V2        | 2022 | arXiv Link  | ~ 4.4 M      |
| EfficientViT b0     | 2023 | arXiv Link  | ~ 2.2 M      |
| EfficientViT b3     | 2023 | arXiv Link  | ~ 46.2 M     |
| MaxViT              | 2022 | arXiv Link  | ~ 119 M      |
| CAFormer            | 2022 | arXiv Link  | ~ 95.8 M     |
| ConvFormer base     | 2022 | arXiv Link  | ~ 96.9 M     |
| ConvFormer medium   | 2022 | arXiv Link  | ~ 54.8 M     |
| DaViT               | 2022 | arXiv Link  | ~ 87.0 M     |
| mViT-v2             | 2021 | arXiv Link  | ~ 50.7 M     |
| Next-ViT            | 2022 | arXiv Link  | ~ 43.8 M     |
| RepViT              | 2023 | arXiv Link  | ~ 7.8 M      |
| RepVGG              | 2021 | arXiv Link  | ~ 86.6 M     |
| XCiT small          | 2021 | arXiv Link  | ~ 25.8 M     |
| XCiT medium         | 2021 | arXiv Link  | ~ 83.8 M     |
| RegNetY 160         | 2020 | arXiv Link  | ~ 80.7 M     |
| RegNetY 320         | 2020 | arXiv Link  | ~ 141 M      |
| MambaOut            | 2024 | arXiv Link  | ~ 81.9 M     |
| BEiTv2              | 2022 | arXiv Link  | ~ 85.3 M     |
| Twins pcvt          | 2021 | arXiv Link  | ~ 43.3 M     |

TABLE II: List of DL-based detectors.

tion, because both of their popularity and because of their uses to address the problem of deepfake detection, see, for instance [9], [10], [33], [34] ; the interested reader is also referred to the survey papers [35], [36].

With a different goal in mind, we included the mobileNet family, as it constitutes the reference lightweight and efficient Convolutional Neural Network (CNN) models. It has been designed for the objective deployment of deep learning object detection and recognition, under the challenging limitations of a resource-constrained environment, such as a microcontroller with limited memory, energy, and power.

We also wanted to include the latest deep learning features extraction and classification models based on vision transformers. These methods indeed have substantially improved the performance of deep learning models in general. Note that these methods have already been widely adopted for the problem of deepfake detection, see, for instance, the survey papers [37], [38]. We included the instance of the well-known

RegNet family because of its popularity, which makes it an unsurprising choice for deepfake detection [20]. Similarly we included the Twins model, as it is based on swin model, which is well known and has already been used for deepfake detection [22], [23].

While we included lightweight models such as mobileNet, we included some representatives of the MetaFormer models [39], namely CAFormer and ConvFormer, as they are popular models as well as examples of rather heavyweight vision transformer models for classification.

With this wide selection, we also wanted to show as much diversity as possible, for example, by including the latest deep learning models as well as older ones, some very light and some very heavy.

### III. NUMERICAL RESULTS AND ANALYSIS

The main results of the present study are presented in Table III. It presents the detection accuracy of all models on the problem of deepfake generator identification. Note

| Name                | Test Accuracy | Epoch duration |
|---------------------|---------------|----------------|
| EfficientNet b0     | 64.38         | 230            |
| EfficientNet b3     | 69.47         | 780            |
| EfficientNet2 m     | 85.30         | 1021           |
| EfficientNet2 s     | 85.21         | 584            |
| MixNet              | 83.89         | 566            |
| NFNet               | 84.31         | 576            |
| ECA-NFNet           | 85.18         | 1393           |
| TinyNet             | 83.87         | 346            |
| ConvNeXt-V2         | 85.34         | 2681           |
| ResNeXt             | 56.35         | 4143           |
| Res2NeXt 50         | 84.17         | 711            |
| ResNet 101          | 84.39         | 854            |
| SEResNext           | 85.08         | 2625           |
| VGG19               | 75.13         | 806            |
| DenseNet            | 84.36         | 834            |
| Xception            | 84.58         | 1604           |
| Inception-V3        | 81.66         | 360            |
| Inception-ResNet-V2 | 83.11         | 725            |
| Lambda ResNet50     | 84.65         | 664            |
| MobileNet-V2        | 83.57         | 323            |
| MobileNet-V3        | 82.36         | 436            |
| MobileNet-V4        | 79.53         | 555            |
| MobileViT-V2        | 84.65         | 463            |
| EfficientViT b0     | 76.58         | 256            |
| EfficientViT b3     | 83.25         | 1100           |
| MaxViT              | 84.13         | 3350           |
| CAFormer            | 85.96         | 3681           |
| ConvFormer base     | 85.44         | 3216           |
| ConvFormer medium   | 85.53         | 6102           |
| DaViT               | 85.23         | 2497           |
| mViT-v2             | 85.09         | 1893           |
| Next-ViT            | 84.78         | 1076           |
| RepViT              | 84.31         | 345            |
| RepVGG              | 83.77         | 1120           |
| XCiT small          | 85.12         | 3131           |
| XCiT medium         | 85.59         | 11429          |
| RegNetY 160         | 84.63         | 1690           |
| RegNetY 320         | 85.59         | 2730           |
| MambaOut            | 86.05         | 2916           |
| BEiTv2              | 76.35         | 1587           |
| Twins               | 83.87         | 964            |

TABLE III: List of DL-based detectors.

that, for speeding up the training process, and in order to get results which are easier to read, the training and the testing was carried out of images of size  $128 \times 128$  and converted into grayscale. The accuracy may seem rather small if one considers that papers on deepfake detection often report detection accuracy is higher than 95%. However, in our case, we trained the model to perform a multi-class classification task with almost 50 classes corresponding each to a generative AI model, see the Table I. Keeping in mind the small image size we used, the detection results are actually very impressive.

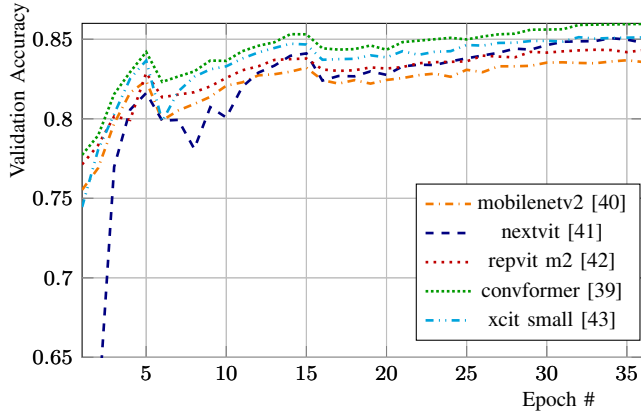
While the number of 36 epochs may seem rather small, we would like to convince the reader that this is enough is our case by presenting in the figure 1 the evolution of the training model. First of all, the figure on the left-hand side shows the evolution of the testing accuracy as a function of the training epoch. On this figure, one can note that a very negligible amount of improvement after epoch 30 even for

the heaviest model, such as ConvFormer and Xcit which are supposedly the longest and the hardest to train.

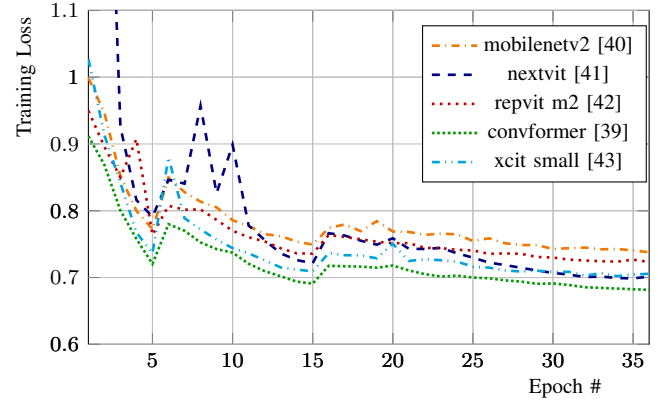
The figure on the right-hand side presents the training loss as a function of the epoch index. Note the very small amount during the third and last cycle, after epoch number 15 and especially during its second hand. The small number of epochs needed to train the models can be explained by large size of the training dataset, which is made up of 670,000 images.

Overall, it is striking to note in Table III that Vision-Transformers generally achieve the best overall results. In other words the five deep learning models that achieve the best performance are all amongst the recent vision-transformer-based models: MambaOut, CAFormer, ConvFormer, RegNet and XCiT.

However, one can also note that, unsurprisingly, that the deep learning models with the highest accuracy are generally amongst the most resource-consuming to train. Alternatively,



(a) Validation loss (Binary Cross-Entropy) over iterative training.



(b) Accuracy (average of true-positive and true-negative rates) over iterative training process.

Fig. 1: Comparison of the accuracy of the proposed model with several of its possible competitors. The test detection accuracy (left) and the validation loss functions (right) are reported during the training process, more precisely plotted as a function of the epoch number.

| average / sum | top-10 | Linear model | non-linear model | Fully connected |
|---------------|--------|--------------|------------------|-----------------|
| 83.93         | 85.51  | 87.27        | 88.69            | 88.72           |

TABLE IV: List of DL-based detectors.

some models seem to be very good choices in terms of the tradeoff between detection accuracy and training resources, especially RepViT, EfficientNet v2 and MobileNet V2 whose training time can be 20 times faster than ConvFormer for only 1% of loss in terms of detection accuracy.

#### A. Ensemble Classifier Methods Using all Deep Learning Models

Last but not least we have tried five different methods for aggregating all the results from all the classifiers in order to further improve the detection accuracy with an ensemble classifier. First we tried as a reference comparison to average the “soft output” all the classifiers. This method is certainly not the most effective, as it gives the same importance to weak classifiers and to the best ones. Nevertheless, we included it in the results as a benchmark to show the improvement in terms of accuracy. Inspired by the work [33] we also tried including the average score of the classifier that gives the largest “soft output”. The idea behind this methodology is to aggregate the score of the  $k$  classifiers who make the most certain decision in favour of a deepfake image. On the opposite, a classifier whose score is so-so shall not be considered.

Then we used three classical methods; first, we trained a linear classifier: with the “soft outputs” from all the models, it computes a weighted sum from all classifiers. The results we present were obtained with a linear SVM, which we implemented using the scikit-learn python package, but others linear classifiers we used gave similar results.

We implemented a non-linear SVM classifier. Even though the number of deep learning models is relatively small, the

relatively high number of images, together with the cross-validation of the hyperparameters of the kernel width and regularization parameter, made the training very long; hence our choice to use a Nyström approximation [44]; empirically, we determined that 200 components was a good tradeoff.

Last but not least, we implemented a 2 layers fully connected classifier with pytorch and we used the well-known Xavier, or Glorot, initialization [45].

As expected, the results presented in the Table IV show that the average of all deep learning models is a poor choice, as it performs worse than many deep learning models alone. Again, this result is not very surprising, as such an aggregation function gives the same importance to all classifiers, the worst as well as the best. The Top- $k$  classifier outputs with  $k = 10$  gave slightly better results. However, we can note that here it is not satisfying at all, as the accuracy of the aggregation of all deep learning models is worse than the best models alone. Interestingly, the linear classifier, though very simple, seems a much better choice, as it improves the accuracy of almost 1.3% which is not a negligible improvement comparing the accuracy of deep learning models. The non-linear SVM classifier, or more precisely its Nyström approximation, as well as the fully connected classifier, allows improving the accuracy further by approximately 1.4%.

Unsurprisingly, the non-linear methods provide the largest improvement in terms of classification accuracy. However, the improvement shall be balanced by the relatively large number of deep learning models we used. Indeed, while a 2.7% improvement in terms of the balanced accuracy is a rather very good result, it is striking that we needed almost

40 deep learning models to reach this result, which is a rather high number of classifiers with regards to the improvement. We can conclude that the classifiers are, most of the time, very much correlated and likely to provide similar decisions.

#### IV. CONCLUSION AND FUTURE WORKS

The present paper proposes a large-scale benchmark for the problem of deepfake detection. Indeed this large scale is based on a wide range of almost 50 different generative AI models used in the training and testing set. Additionally, the paper proposes a numerical comparison of over 40 models for deepfake detection ranging from the simplest and oldest ones, such as DenseNet, ResNet and VGG because they are popular and often used in deepfake detection, up to the latest vision-transformers based deep learning classification models. We show that, generally speaking, vision transformers perform extremely well for this task of weak signal detection. Additionally, we also demonstrate empirically that larger models provide the best overall performance. Nevertheless some models are good tradeoffs in terms of deepfake detection accuracy and training complexity, such as EfficientNet v2 and MobileViT. We are strongly believed that the study presented in the present paper will serve for the design of future deepfake detection methods and will become a reference benchmark for the evaluation of future deep learning methods.

Upon acceptance of the present paper, we will release the dataset of deepfake images along with the weights of the trained models to further serve future research in the field of deepfake detection.

We also believe that our large dataset of deepfake images can be used to study the problem of generalization when testing over out-of-distribution images.

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